A Neural Network Method for Land Surface Temperature and Emissivity Retrieval From Hyperspectral Thermal Infrared Data by Incorporating Prior Surface Knowledge

Xiu-Juan Li, Li Ni, Yuan-Liang Cheng, Lei Lei, and Xing-Xing Zhang

***Abstract*—Land surface temperature (LST) and land surface emissivity (LSE) are the core parameters in thermal infrared (TIR) remote sensing research, which play crucial roles in surface energy balance, climate change studies, vegetation monitoring, and agricultural management. With advances in remote sensing sensors, hyperspectral TIR (HTIR) remote sensing has created new possibilities for LST and LSE retrieval. However, the rich spectral information provided by HTIR data remains underutilized. To address this, this study proposes a neural network method by incorporating prior surface knowledge for the simultaneous retrieval of LST, LSE, atmospheric transmittivity, upwelling radiance, and downwelling radiance. The method enhanced input features by incorporating prior surface knowledge with a split-window (SW)-like method and reduced the dimensionality of output space with principal component analysis (PCA), thereby improving the efficiency of the artificial neural network (ANN) retrieval process. Results demonstrated that an RMSE of 1.5 K for LST and 0.011 for LSE were achieved by the proposed method, while also improving the spectral shape rationality of LSE. Moreover, cross-validation further confirmed that the retrieval results aligned well with existing products. Overall, this study successfully improves the information efficiency of machine learning (ML) models, providing a promising technical framework for the land surface and atmospheric parameter estimation from HTIR remote sensing data.**

***Index Terms*—Emissivity, hyperspectral, land surface temperature (LST), machine learning (ML), split window-like method, thermal infrared (TIR) remote sensing.**

I. Introduction

LAND surface temperature (LST) and emissivity the core parameters in thermal infrared (TIR) remote sensing [1], [2], which are widely applied in surface energy balance analysis [3], [4], [5], [6], climate change research [7], [8], [9], ecological [10], [11], [12], [13] and hydrological monitoring [14], [15], [16], and so on. After decades of development, retrieval methods based on multispectral TIR remote sensing have matured and been widely adopted in practical applications [2]. However, multispectral TIR sensors are inherently constrained by their limited spectral bands, which typically necessitates the stepwise estimation of land surface and atmospheric variables. The stepwise methods usually rely on a series of specific assumptions. For instance, atmospheric parameters and land surface emissivity (LSE) are known a priori [3], [17], [18], [19]. Such assumptions not only restrict the applicability of the method but also lead to error accumulation during the retrieval process. Consequently, further improvements are challenging to achieve, especially in scenarios with high-water vapor content or complex surface topography.

Hyperspectral TIR (HTIR) remote sensing has offered novel pathways for land surface and atmospheric parameter retrieval [2], [20], [21], [22]. Specifically, its continuous and rich information may impose more reasonable physical constraints on radiative transfer processes, which may further enhance the accuracy and efficiency of surface parameter retrieval [23], [24], [25]. Currently, how to fully leverage the advantages of hyperspectral data for LST retrieval has become a critical issue that demands urgent resolution [1], [2], [23], [24], [25], [26].

Machine learning (ML) techniques have been progressively introduced into the retrieval of TIR parameters, owing to their inherent advantages in addressing high-dimensional, nonlinear, and ill-posed problems. Methods such as artificial neural networks (ANNs) have been widely applied to the retrieval of LST, atmospheric profiles, and water vapor content [27], [28]. Driven by the fast progress of deep learning technologies, convolutional neural networks [29], graph neural networks [30], and transfer learning models [31] have also been preliminarily explored. Typically, these methods take top-of-atmosphere (TOA) radiance as input and land surface or atmospheric parameters as output, establishing end-to-end retrieval models through training on simulated datasets.

However, current research using ML techniques primarily focuses on retrieving individual parameters, while the interactions between these parameters have received relatively little attention. Furthermore, ML methods often fail to fully

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capture the continuity and correlations across different channels, which results in spectral shapes deviated from physical reality, thereby limiting the practical application potential [32], [33], [34]. Additionally, the land surface and atmospheric parameter retrieval is inherently challenging due to their strong interdependence. For data-driven ML methods, how to provide substantial and reliable information is crucial [34], [35], [36], [37].

On the one hand, the information redundancy of hyperspectral data needs to be reduced. Although HTIR data contain numerous channels with abundant information, strong correlations between channels can induce information redundancy, potentially aggravating the ill-posed retrieval problem [38]. Therefore, it is essential to balance information richness with model stability through appropriate channel selection and dimensionality reduction for ML methods. Particularly for LSE, the ability to accurately reconstruct its spectra [39] directly affects the feasibility of subsequent applications. Given the distinctive characteristics of HTIR data, developing effective strategies for spectral information extraction remains an important research direction [36], [40].

Meanwhile, ML models rely heavily on the information content of the input data. Existing models typically use only TOA radiance as input, which offers limited information in scope. Enriching the input with prior knowledge or derived surface radiance information may potentially enhance the retrieval accuracy. Specifically, obtaining surface radiance requires atmospheric correction, a process that usually involves radiative transfer models and atmospheric profiles. This procedure is not only complex but also constrained by the accuracy of the profiles, limiting its general applicability. In this context, Zheng et al. [41] proposed a new method for atmospheric correction that directly retrieves ground-leaving radiance by exploiting differences in atmospheric absorption across spectral bands. It offers an efficient pathway for acquiring land surface information. However, its performance on HTIR data has yet to be fully evaluated.

To address these challenges, this study developed a neural network method by incorporating prior surface knowledge for HTIR data. This method uses the SW-like method and principal component analysis (PCA) to provide essential knowledge and reduce model complexity for the ANN-based retrieval, thereby carrying out efficient retrieval of land surface and atmosphere parameters. The sections of this article are structured as follows. Section II presents the datasets utilized in this study. Section III details the technical methodology of the proposed model. The results and validation of the method are reported in Section IV. Section V presents the discussion, and Section VI summarizes the conclusion.

II. Data

A. Simulated Dataset

In this study, a simulated database was established for building the retrieval model of land surface and atmosphere parameters. Data simulation was primarily constructed using the Thermodynamic Initial Guess Retrieval (TIGR) database, the ECOsystem Spaceborne Thermal Radiometer Experiment on Space Station (ECOSTRESS) spectral library, and the Operational release for Automatized Atmospheric Absorption Atlas (4A/OP) model.

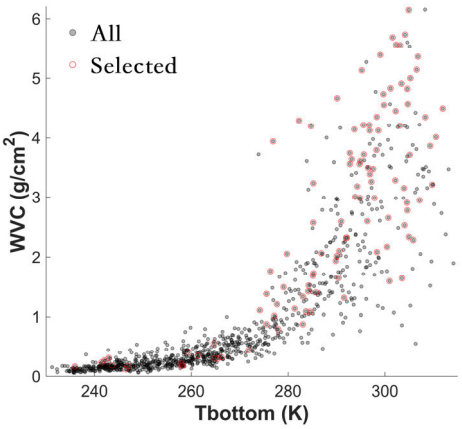


Fig. 1. Selected atmospheric profiles from the TIGR database.

The TIGR database used in this study was released in May 2010, providing 2311 atmospheric profiles that encompass diverse global atmospheric conditions. In total, 110 atmospheric profiles were selected from the database to ensure both global representativeness and a uniform distribution of water vapor conditions. The selection criteria were based on two primary considerations. First, only cloud-free atmospheric profiles were retained to guarantee the validity of radiative transfer simulations. Second, to prevent an overrepresentation of profiles within low-humidity intervals, a stratified random sampling approach was employed. Specifically, multiple profiles were randomly selected within each humidity interval, thereby achieving a more balanced distribution across varying humidity conditions. Fig. 1 illustrates the distribution of water vapor content and bottom-of-atmosphere temperature (T_{bottom}) across the selected atmospheric profiles. Subsequently, the 4A/OP model was employed to simulate atmospheric transmittivity, upwelling radiance, and downwelling radiance with the selected profiles. Furthermore, channel response convolution was performed on these simulated atmospheric parameters using the response functions of the Cross-track Infrared Sounder (CrIS), yielding corresponding simulated HTIR data.

Simultaneously, LSE spectra of typical land surfaces were extracted from the ECOSTRESS spectral library. Its V1.0 version was released in 2018, covering over 3400 natural and artificial surfaces, including more than 1100 spectra of vegetation and nonphotosynthetic vegetation. To ensure the selected emissivity spectra represent diverse land surface conditions, 102 spectra were chosen for data simulation, encompassing 18 rock types, 36 soil types, 12 artificial surfaces, 30 vegetation types, and 6 water bodies. The spectral characteristics of the selected spectra are illustrated in Fig. 2.

LST was treated as a disturbance variable to ensure that the simulated data possessed satisfactory representativeness, with its values determined based on T_{bottom}. Specifically, considering the correlation between LST and T_{bottom}, the LST was set within a range of −20 to +20 K relative to T_{bottom}, with

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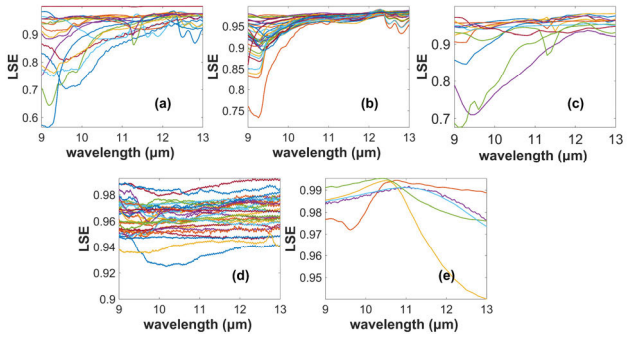


Fig. 2. Selected emissivity spectra of different land covers. (a) Rock. (b) Soil. (c) Man-made. (d) Vegetation. (e) Water.

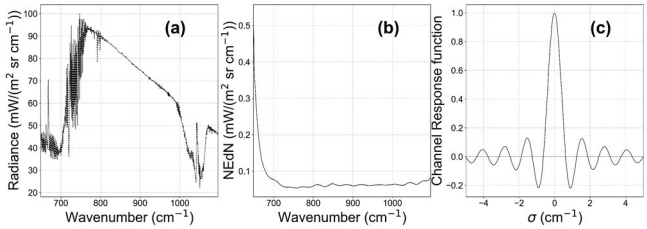


Fig. 3. Sample of (a) radiance, (b) NEdN, and (c) channel response function of CrIS data. (From CrIS data on February 2, 2023, FOV5).

a step size of 5 K. The simulated LST range falls between 220 and 330 K, which aligns well with the actual thermal conditions of natural land surfaces. Subsequently, by integrating the simulated atmospheric parameters, LST values, and the selected emissivity spectra into the TIR radiative transfer model, the simulated radiance datasets for HTIR sensors were generated. Ultimately, these calculations yielded a total of 100 980 sample pairs, encompassing 9 LST conditions, 102 surface emissivity spectra, and 110 atmospheric profile types.

B. CrIS Data

The satellite-based HTIR data used in this study were acquired from the CrIS, which is installed on the Suomi National Polar-orbiting Partnership (S-NPP) satellite that lifted off on October 28, 2011.

As a Fourier transform infrared spectrometer, the CrIS sensor covers a spectral range of 3.9–15.4 μm, which includes 713 TIR channels (specifically within the 9.14–15.38 μm) and features a spatial resolution of approximately 14 km. Notably, the spectral resolution of its TIR channels is 0.625 cm^{-1}. Its corresponding channel response functions are presented in Fig. 3(c). For this study, we utilized its 523 TIR channels that fall within 9–13 μm.

Currently, the Level 1B (L1B) products of CrIS provide TOA radiance that has undergone calibration, enabling land surface and atmospheric parameter retrieval. Furthermore, these L1B products include supplementary datasets such as noise equivalent differential radiance (NEdN), observation angles, and quality control metrics. Detailed information is publicly available at the official National Oceanic and Atmospheric Administration (NOAA) repository: https://www.star.nesdis.noaa.gov/jpss/CrIS.php. Examples of the CrIS TOA radiance and corresponding NEdN datasets are illustrated in Fig. 3.

C. Products Used for Cross-Validation

1) VIIRS LST Product: The Visible Infrared Imaging Radiometer Suite (VIIRS) and CrIS sensors are mounted on the same satellite platform, providing an excellent comparative reference. The VIIRS sensor features 22 spectral channels covering a wavelength range of 0.41–12.01 μm, encompassing nine visible and near-infrared (VNIR) channels, one panchromatic channel, eight midinfrared (MIR) channels, and four TIR channels. It offers a nominal spatial resolution of 750 m and a 16-day revisit period. For comprehensive technical specifications, refer to the official documentation: https://ladsweb.modaps.eosdis.nasa.gov/missions-and-measurements/viirs/

Notably, VIIRS provides a standard LST product retrieved from channels 15 (10.76 μm) and 16 (12.01 μm) via the split-window (SW) method (Schueler et al. [42]). Validation analyses have shown that the VIIRS LST product can achieve an accuracy of within 3 K during daytime and within 1.5 K during nighttime [43], [44]. Since VIIRS and CrIS are deployed on the same satellite, the temporal discrepancy between their observations can be negligible [45]. Therefore, the VIIRS LST product was adopted as the reference dataset to validate the LST retrieval accuracy of the proposed method in this study.

2) AIRS LSE Product: The Atmospheric Infrared Sounder (AIRS) has been aboard NASA's Aqua satellite since 2002. It features 2378 infrared channels and 4 VNIR channels, covering a spectral range of 0.4–1.0 and 3.7–15.4 μm. Designed specifically to acquire high-accuracy atmospheric temperature profiles and other land surface and atmosphere products, AIRS offers a horizontal spatial resolution of 13.5 km at nadir and a spectral resolution of ~0.55 cm^{-1} in the TIR bands. Detailed information is available at https://aqua.nasa.gov/content/airs

Notably, AIRS Level 2 (L2) products provide LSE across multiple TIR bands [46]. Previous validation studies have assessed the reliability of these AIRS LSE products using Advanced Spaceborne Thermal Emission and Reflection Radiometer (ASTER) data and laboratory measurements. Specifically, the validation yielded absolute errors of 2.32% and 2.26% over the Namib Desert, and 0.68% and 2.26% over the Kalahari Desert, respectively [47]. These findings illustrate that the AIRS LSE product demonstrates robust accuracy, thereby making it a suitable reference dataset for validating the LSE retrieval results of the proposed method in this study.

III. Methodology

In this study, a neural network method was proposed for LST and LSE retrieval using HTIR data. Specifically, the method enriches the input information for the ANN model by incorporating the SW-like method, which provides the leaving-ground brightness temperature as prior surface knowledge for the retrieval model. Moreover, the integration of PCA helps reduce the model complexity. This integrated framework enables the direct retrieval of LST, LSE, atmospheric transmissivity, upwelling radiance, and downwelling radiance from

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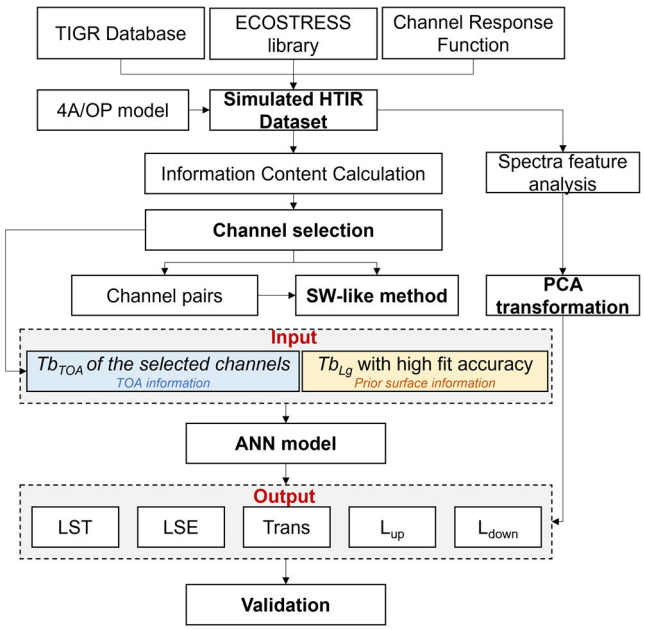


Fig. 4. Flowchart of land surface and atmospheric parameter estimation with HTIR data.

HTIR observations. As shown in Fig. 4, the workflow of the proposed method is illustrated.

A. Channel Selection

Channel selection aimed to ensure that the utilized data could deliver the maximum amount of surface and atmospheric information. A stepwise iterative method based on information content is utilized in this study to determine the optimal combination of channels. This method comprehensively considers channel weights, observational noise, and channel correlation, which is widely recognized as one of the most effective channel selection approaches [48].

Information content is determined by the difference in information entropy of the preobservation and postobservation states

$$H = \frac{1}{2}\ln|S_a| - \frac{1}{2}\ln\left|\hat{S}\right| = -\frac{1}{2}\ln\left|\hat{S}S_a^{-1}\right| \tag{1}$$

where S_a denotes the background covariance matrix, which can be acquired from the average LST and LSE values for different climatic conditions and land-cover types calculated using simulated data. $\hat{S}$ represents the postobservation covariance matrix, which can be calculated based on S_a

$$\hat{S} = S_a - S_a K^T \left(KS_aK^T + S_e\right)^{-1} KS_a \tag{2}$$

where K is the weighting function matrix, which can be obtained by computing the partial derivatives of the TOA brightness temperature with respect to surface and atmospheric parameters. S_e denotes the observation error covariance matrix. It is a diagonal matrix in which every diagonal element corresponds to the covariance of the observation error.

To reduce computational complexity, a stepwise iterative approach was employed for channel selection [49], [50]. Let k denote the row vector of the weight function matrix K corresponding to the weight function of a specified channel. Let s_e denote a diagonal element in the S_e matrix that represents the observation error variance for the same channel. For this specific channel, $\hat{S}$, the error covariance matrix derived after observation can be written as follows:

$$\hat{S} = S_a - S_a k^T \left(kS_ak^T + s_e\right)^{-1} kS_a \tag{3}$$

where $kS_ak^T + s_e$ is a single value. Subsequently, we can derive the formula that applies to the calculation of the information content of a single channel

$$h = -\frac{1}{2}\ln\left|I - \frac{(kS_a)^T k}{s_e + k(kS_a)^T}\right|. \tag{4}$$

According to the above equation, the channel corresponding to the maximum h value would be selected as the optimal channel for the current iteration. For each iteration, once a channel is chosen, the selected channel is used to update $\hat{S}$, which then acts as S_a in the next round of iteration. Thereafter, the unselected channels in the current iteration become candidates for subsequent iterations. In this manner, the already selected channels are incorporated into the background information. The information content of the remaining candidate channels is then recalculated based on the updated background covariance matrix, thereby reducing the interchannel correlation among the selected channels. This process is repeated until channels sensitive to either land surface or atmospheric information are identified.

B. SW-Like Method for Tb_{Lg} Obtaining

Building upon the SW method, Zheng et al. [41], [51] proposed an SW-like method that can directly retrieve leaving-ground brightness temperature without relying on external atmospheric information. The formula for the algorithm is presented as follows:

$$Tb_{Lg_i} = a_0 + a_1T_i + a_2\left(T_i - T_j\right) + a_3\left(T_i - T_j\right)^2 \tag{5}$$

where $a_i (i = 0, \ldots, 3)$ represent the coefficients of the SW-like method. T_i and T_j are the TOA brightness temperature of channel i and j, respectively.

Based on the simulated dataset, channels i and j are selected using an iterative approach. For each channel i, the remaining channels are iteratively considered as channel j. Each candidate channel j was then substituted into the SW-like method to estimate the leaving-ground brightness temperature corresponding to channel i. The retrieval results were compared with the true values, and valid channel pairs (i, j) were selected based on a predefined error threshold. Consequently, the leaving-ground brightness temperatures for all HTIR channels can be derived.

C. PCA Transformation

Given the strong correlations among hyperspectral channels, PCA transformation was employed to extract essential spectral information and reduce the output dimension of the model. PCA is a widely used data preprocessing method in hyperspectral retrieval studies [52], [53]. It projects data onto a new coordinate system, enabling the projected data to capture the main characteristics of the original dataset [54].

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The resulting variables, each called a principal component, are linear combinations of the original variables. Where the first principal component represents the largest proportion of variance, with subsequent components explaining progressively less. By applying PCA, multivariate data can be effectively compressed and represented in a lower-dimensional space. The transformation is expressed as follows:

$$Y_m = X \cdot W_m \tag{6}$$

where Y_m denotes the first m PCs of the spectra. X is the data matrix of LSE and atmospheric parameters. W_m is the weight matrix consisting of the eigenvectors corresponding to the m largest eigenvalues of $X^T X$.

In this study, spectral feature analysis was conducted to establish the optimal number of principal components by assessing the impact of principal component scores on spectral reconstruction errors. Ultimately, 20 principal components were retained for emissivity, while ten principal components were used for each of the three atmospheric parameters.

D. ANN Model Building

For HTIR data, this study employed the ANN method to retrieve LST, LSE, atmospheric transmittivity, upwelling radiance, and downwelling radiance. However, due to the deep coupling between surface and atmospheric parameters, their simultaneous retrieval requires sufficient information support. Therefore, the leaving-ground brightness temperatures derived from the SW-like method were used as prior knowledge to input into the ANN model. Furthermore, PCA was applied to reduce data redundancy by transforming the spectral data of LSE, atmospheric transmittivity, upwelling radiance, and downwelling radiance. Ultimately, the ANN model was constructed based on the simulated dataset. The inputs of the ANN model include the leaving-ground brightness temperatures that meet accuracy requirements and the TOA radiance corresponding to the selected channels. While the outputs consist of LST, 50 principal components of LSE, transmittivity, upwelling radiance, and downwelling radiance. After hyperparameter debugging, the model was set up with two hidden layers, each containing 170 nodes

$$LST, Y_{\varepsilon_n}, Y_{\tau_n}, Y_{L_n^{\uparrow}}, Y_{L_n^{\downarrow}} = f\left(Tb_{Lg_{1,\ldots,m}}, L_{1,\ldots,n}\right) \tag{7}$$

where $Tb_{Lg_{1,\ldots,m}}$ represents the leaving-ground brightness temperatures of m channels that meet accuracy requirements. $L_{1,\ldots,n}$ denotes the TOA radiance of n selected channels. $Y_{\varepsilon_n}, Y_{\tau_n}, Y_{L_n^{\uparrow}}, Y_{L_n^{\downarrow}}$ are the principal components of the emissivity and atmospheric parameters for the n selected channels, respectively. The spectral characteristics of emissivity and atmospheric parameters can be recovered through the inverse PCA transformation

$$\begin{aligned} \varepsilon_n &= Y_{\varepsilon_n} W_{\varepsilon_n}^T \\ \tau_n &= Y_{\tau_n} W_{\tau_n}^T \\ L_n^{\uparrow} &= Y_{L_n^{\uparrow}} W_{L_n^{\uparrow}}^T \\ L_n^{\downarrow} &= Y_{L_n^{\downarrow}} W_{L_n^{\downarrow}}^T \end{aligned} \tag{8}$$

where $W_{\varepsilon_n}^T, W_{\tau_n}^T, W_{L_n^{\uparrow}}^T, W_{L_n^{\downarrow}}^T$ are the corresponding principal component coefficients.

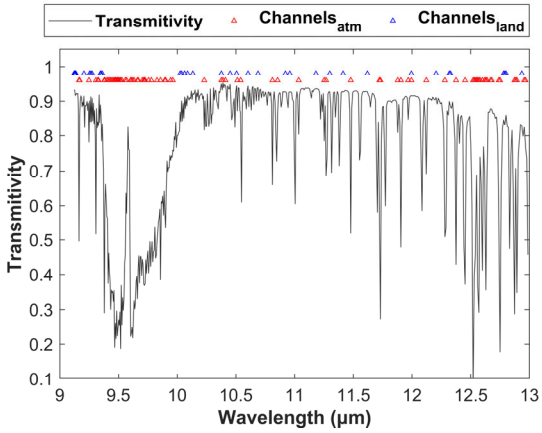


Fig. 5. Position of selected channels that are sensitive to land surface and atmospheric information.

IV. Result

A. Result of Channel Selection

To support land surface and atmospheric parameter retrieval, this study adopted the information content-based stepwise iterative method for feature channel selection. For channels sensitive to different parameters, the surface and atmospheric information content of each channel are separately quantified using the stepwise iterative method. For surface-sensitive channels, the number of selected channels is determined based on our previous study [52], following a targeted evaluation of their information content, resulting in a total of 35 channels. For atmosphere-sensitive channels, those with an information content exceeding 70% are selected after computing their sensitivity to atmospheric parameters, yielding a total of 85 channels. After removing two duplicate channels, a total of 118 feature channels were selected, accounting for less than one-quarter of the original 523 channels. The spectral locations of these selected channels are presented in Fig. 5. Consequently, the 118 selected feature channels can be used for the retrieval of land surface and atmosphere parameters. This process simplified the complexity of the ANN model while retaining comprehensive information with minimal redundancy.

B. Accuracy of SW-Like Method

Leaving-ground brightness temperatures were obtained using the SW-like method, thereby providing surface radiation information for the ANN model. The retrieval accuracy of leaving-ground brightness temperatures derived from the SW-like method is presented in Fig. 6. It can be observed that the accuracy was strongly associated with atmospheric absorption characteristics. Accuracy was lower at positions with low atmospheric transmittivity, while it is significantly higher in atmospheric window regions. Specifically, the higher-accuracy regions are primarily concentrated within the atmospheric window zones of 9–9.5 and 10–11 μm. Among the 118 selected channels, 9 brightness temperatures across all 118 channels was 2.65 K.

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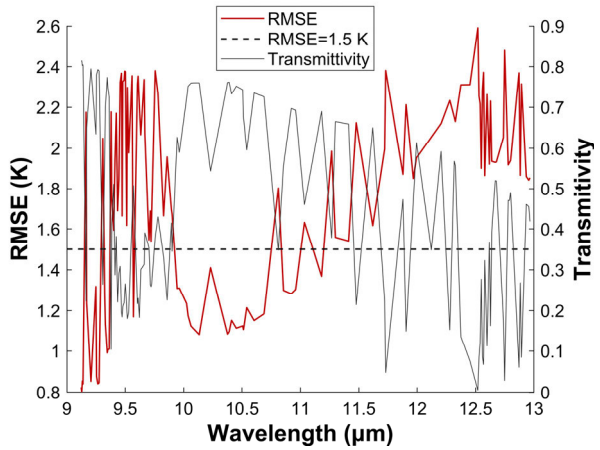


Fig. 6. Accuracy of the SW-like method.

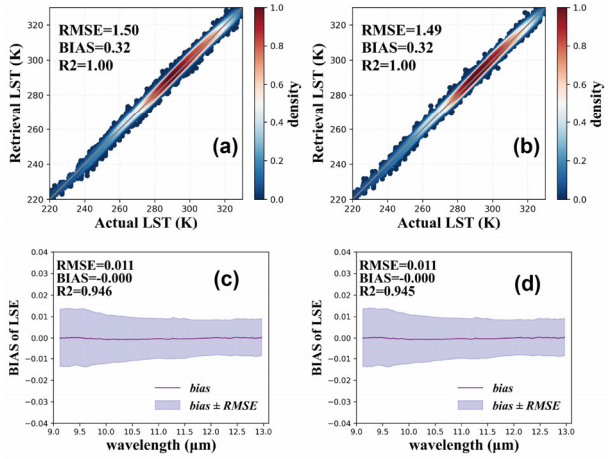


Fig. 7. Retrieval accuracy of LSE and LSE. (a) Result of LST in the training set. (b) Result of LST in the test set. (c) Result of LSE in the training set. (d) Result of LSE in the test set.

C. Results of the ANN Model

After completing channel selection and prior knowledge estimation, this study constructed an ANN model using a simulated dataset. Notably, the LSE spectra of the 118 selected channels were transformed into 20 principal components, while the three atmospheric parameters were transformed into 10 principal components, respectively. Therefore, the inputs of the ANN model comprised the TOA radiance corresponding to the 118 selected channels and the leaving-ground brightness temperatures that met accuracy requirements (taking the results with an RMSE < 1 K as an example). The outputs included LST and 50 principal components. Subsequently, continuous spectra of LSE and atmospheric parameters can be obtained using the PCA inverse transformation.

Fig. 7 shows the retrieval results for surface parameters. For LST, the RMSE for both the training and test datasets reached 1.5 K. This consistent accuracy level not only verified the reliability of the retrieval results but also revealed that the ANN model possessed favorable stability and robustness. Regarding LSE, the RMSE of the two datasets was uniformly 0.011, indicating high retrieval accuracy. However, relatively

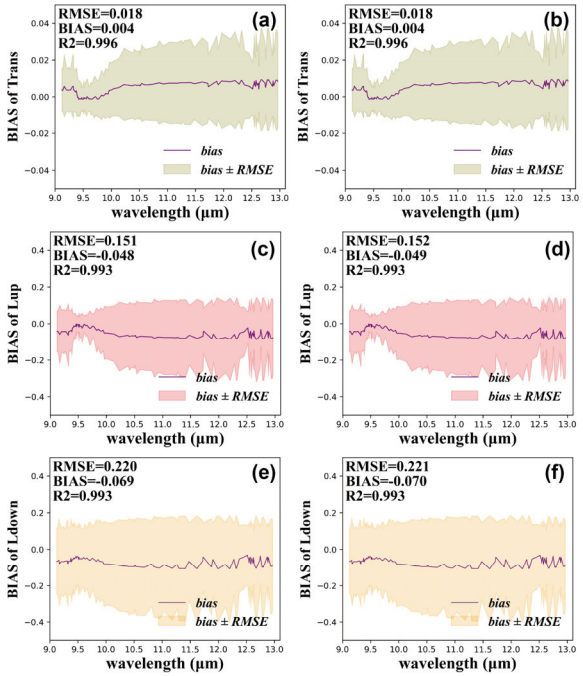


Fig. 8. Retrieval accuracy of atmospheric transmittivity, upwelling radiance, and downwelling radiance. (a) Result of atmospheric transmittivity in the training set. (b) Result of transmittivity in the test set. (c) Result of atmospheric upwelling radiance in the training set. (d) Result of atmospheric upwelling radiance in the test set. (e) Result of atmospheric downwelling radiance in the training set. (f) Result of atmospheric downwelling radiance in the test set.

larger errors were observed in the 9–10-μm range, which may be attributed to the inherently high variability across different land cover types within this range, especially for soil and rock surfaces.

Fig. 8 presents the retrieval results for three atmospheric parameters. As a result, the test set showed an RMSE of 0.018 for atmospheric transmittivity, 0.152 W cm^{-2} μm^{-1} sr^{-1} for atmospheric upwelling radiance, and 0.221 W cm^{-2} μm^{-1} sr^{-1} for atmospheric downwelling radiance. Overall, the retrieval accuracy of atmospheric parameters was generally satisfactory. However, compared to the results for LSE spectra, the RMSE of atmospheric parameters exhibited greater variability across different spectral bands. This may stem from the fact that the spectral curves of atmospheric parameters feature numerous absorption peaks and troughs. Consequently, their shapes are more fluctuating, which increases the difficulty of achieving accurate estimation of atmospheric parameters.

D. Sensitivity Analysis

To conduct a further analysis of how noise equivalent temperature difference (NEΔT) affects model accuracy, five levels of random noise were introduced, ranging from 0 to 0.5 K in increments of 0.1 K. This range was defined based on the typical noise characteristics of CrIS, for which the NEΔT generally does not exceed 0.5 K, as illustrated in Fig. 3(c). Specifically, the TOA brightness temperature with different random noises was used for the retrieval of land surface and atmospheric parameters, and the resulting changes in retrieval accuracy were analyzed.

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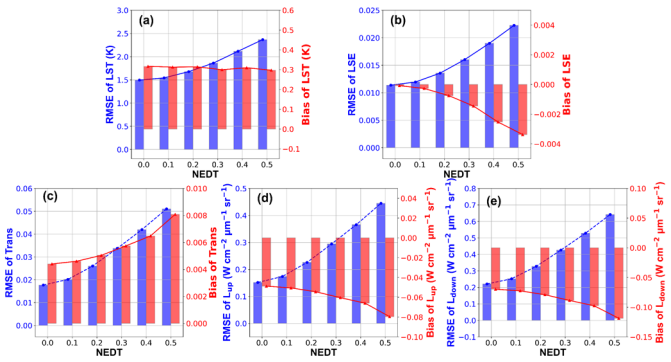


Fig. 9. Sensitivity to NEΔT for the ANN model. (a) LST. (b) LSE. (c) Atmospheric transmittivity. (d) Atmospheric upwelling radiance. (e) Atmospheric downwelling radiance.

Fig. 9 presents the variation of retrieval accuracy with NEΔT for land surface and atmospheric parameters derived from the proposed method. As illustrated in the figure, the accuracy of surface and atmospheric parameters gradually decreased with increasing NEΔT, with a notable acceleration in the decline rate. This trend may be attributed to the increased errors in leaving-ground brightness temperature caused by rising NEΔT. Subsequently, these errors accumulated in the input data, which in turn accelerated the degradation of the retrieval accuracy. Overall, within the NEΔT range of the CrIS instrument, the retrieval accuracy of all parameters showed degradation as instrumental noise increased, while still maintaining satisfactory performance.

E. Application to CrIS Data

To validate the proposed method on actual satellite observations, retrieval experiments were conducted using CrIS data from May 16, 2023. The Iberian Peninsula and its surrounding regions constitute the study area in this work. This region is characterized by predominantly clear-sky conditions throughout the year, with limited cloud cover and low precipitation, making it well-suited for remote sensing applications. Given the relatively coarse spatial resolution of CrIS, cross-validation was performed against the VIIRS LST product and the AIRS LSE product.

Since CrIS and VIIRS are onboard the same satellite platform, their acquisition times can be considered simultaneous. To account for the difference in spatial resolution, outliers were first removed from the VIIRS imagery, followed by spatial aggregation to match the CrIS scale. The retrieved LST was then compared with the aggregated VIIRS product, as illustrated in Fig. 10.

The results showed strong consistency in spatial distribution between the two datasets. Additionally, their value ranges were comparable, with an RMSE reaching 3.65 K. In comparison with the VIIRS product, the retrieved LST values were more clustered, with almost no values below 280 K. This discrepancy may be attributed to the loss of detail caused by significant mixed-pixel effects at the coarser resolution.

For LSE, we compared the retrieved results with AIRS LSE products. Since AIRS and CrIS operate on different satellites,

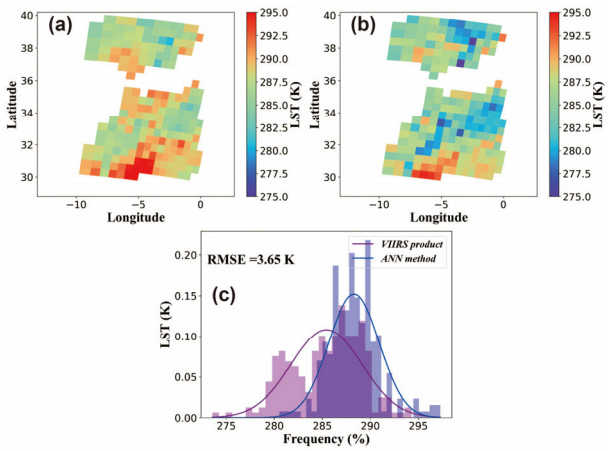


Fig. 10. Comparison of LSTs obtained from the proposed method and VIIRS LST product. (a) Retrieved LST. (b) VIIRS LST product. (c) Comparison between the two.

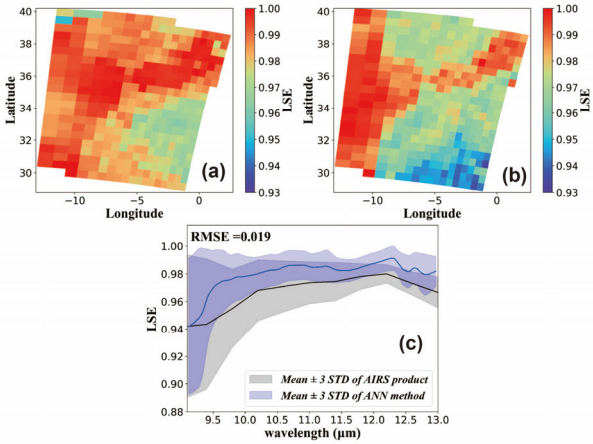


Fig. 11. Comparison of LSEs retrieved using the proposed method and AIRS LSE product (11.0 μm). (a) Retrieved LSE. (b) AIRS LSE product. (c) Comparison between the two.

both temporal and spatial matching were required. Temporally, considering the stability of emissivity, AIRS data acquired within 1 h of the CrIS overpass were selected. Spatially, the AIRS product was resampled to match the spatial resolution of CrIS. In addition, since the AIRS emissivity product is only available for selected TIR channels, the retrieved LSEs were interpolated to the corresponding AIRS spectral channels for comparison.

The comparison results are depicted in Fig. 11. As can be seen from the figure, both datasets showed broadly similar spatial distributions. However, the retrieved emissivity values tend to be systematically higher. The pattern is also observed across other spectral bands [see Fig. 11(c)]. Despite this offset, the spectral shapes of the two LSE datasets are consistent, with an overall RMSE of 0.019, indicating a high degree of similarity.

V. Discussion

A. Impact of the SW-Like Method on Model Accuracy

To evaluate how incorporating the SW-like retrieval results influences model performance, this study examined the effect

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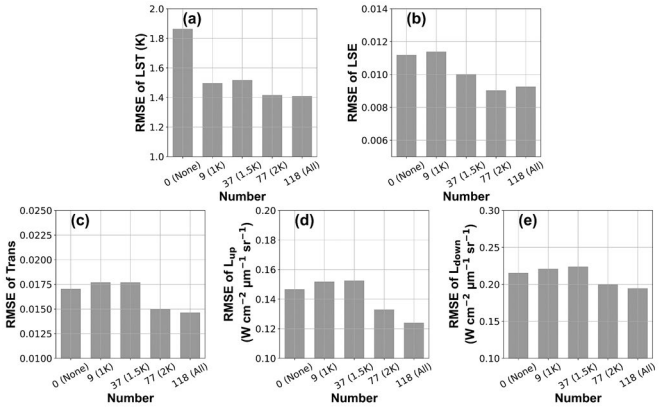


Fig. 12. Influence of knowledge number and accuracy on method performance. (a) LST. (b) LSE. (c) Atmospheric transmittivity. (d) Atmospheric upwelling radiance. (e) Atmospheric downwelling radiance.

of adding ground-leaving brightness temperatures from different channel sets on the accuracy of the ANN models. Based on the retrieval accuracy of the SW-like method, brightness temperatures from four channel groups were sequentially added as prior knowledge to the model input: 9 channels (accuracy <1 K), 37 channels (<1.5 K), 77 channels (<2.0 K), and all 118 channels (accuracy 2.65 K). The resulting changes in retrieval accuracy for surface and atmospheric parameters were then analyzed.

The influence of the number of prior-knowledge channels on the accuracy of retrieving surface and atmospheric parameters is shown in Fig. 12. For LST and LSE, the incorporation of prior knowledge enhanced retrieval accuracy on the whole. Especially for LST, the accuracy in all cases had been improved by more than 0.3 K. This indicated that the incorporation of prior knowledge about the surface had assisted in the retrieval of surface parameters. In contrast, the retrieval accuracy for the spectra of atmospheric parameters fluctuated, with the best performance achieved when all 118 channels were included. This may be attributed to the fact that LST and LSE exhibit greater sensitivity to prior surface information. Therefore, the incorporation of prior knowledge effectively constrained the solution space and improved the retrieval accuracy of surface parameters. In contrast, atmospheric parameters exhibit a weaker dependence on such surface priors and rely more strongly on the full spectral information for accurate characterization. As a result, the introduction of surface prior information did not significantly improve atmospheric retrievals and may even introduce additional uncertainty, leading to slight fluctuations in their retrieval accuracy. Notably, as discussed in Section IV-D, while using more channels provides richer information, it also introduces greater errors. Therefore, incorporating prior surface knowledge requires an optimal balance between model complexity and information content.

B. Impact of PCA for Spectral Reconstruction

To evaluate the impact of PCA transformation on model performance, this study compared the accuracy of parameter outputs directly by the model with those reconstructed after PCA transformation. The comparison results are summarized in Table I.

TABLE I
INFLUENCE OF PCA TRANSFORMATION ON LSE SPECTRA ACCURACY

Parameters \ Method	With PCA	Without PCA
LST (K)	1.49	1.56
LSE	0.011	0.012
Trans	0.018	0.025
Lup (W cm^{-2} sr^{-1} um^{-1})	0.152	0.213
Ldown (W cm^{-2} sr^{-1} um^{-1})	0.221	0.325

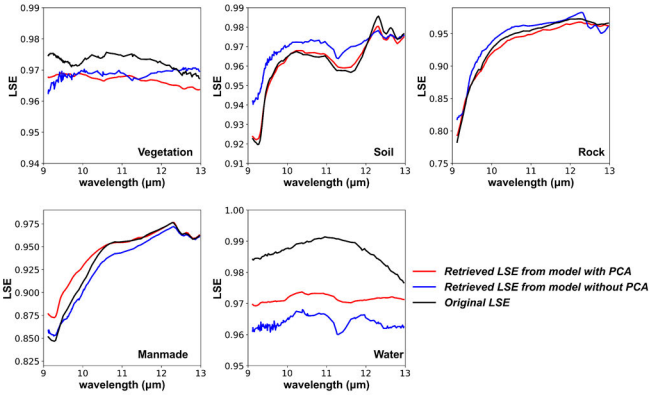


Fig. 13. Influence of PCA transformation on LSE spectra accuracy.

The results indicated that PCA transformation generally improved the retrieval accuracy of all parameters. Specifically, the RMSE of LST decreased by 0.07 K, while LSE accuracy improved by 0.001. Transmittivity retrieval showed an increase of 0.007 in accuracy, and the atmospheric upwelling and downwelling radiance improved by 0.061 and 0.104 W cm^{-2} sr^{-1} μm^{-1}, respectively. Moreover, as illustrated in Fig. 13, the spectral shape of emissivity aligned more closely with actual conditions after PCA transformation. These findings demonstrated that incorporating PCA transformation effectively enhanced both the accuracy in retrieval and spectral fidelity of land surface and atmospheric parameters.

C. SW-ANN Framework for LST and LSE Retrieval

Over the past few years, the SW-TES method has been widely applied in multispectral TIR remote sensing [51], [55], [56], [57], [58]. Inspired by these methods, this study proposes an SW-ANN framework, in which the leaving-ground brightness temperatures derived from the SW-like method are used as inputs to an ANN model to retrieve LST and LSE. After hyperparameter tuning, the network was configured with two hidden layers, each containing 150 nodes, forming the final SW-ANN model.

The results are presented in Fig. 14. Under noise-free conditions, the SW-ANN model achieved satisfactory performance, with an LST accuracy of 1.4 K and an emissivity accuracy of 0.010, slightly outperforming the ANN model

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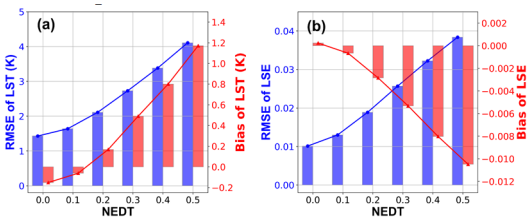


Fig. 14. Sensitivity to NEΔT for the SW-ANN framework. (a) LST. (b) LSE.

incorporating prior surface knowledge developed in this study. However, as the NEΔT increases, the retrieval accuracy of the SW-ANN model degraded rapidly, with a much faster degradation rate than that of the ANN model incorporating prior surface knowledge (see Fig. 9). This may be attributed to the accumulation of NEΔT across hyperspectral channels, which propagated through the sequential SW and ANN processes, leading to amplified errors. In contrast, the ANN model developed in this study incorporated both TOA and leaving-ground brightness temperatures as inputs, which reduced the reliance on the SW-like method alone, thereby providing more reliable information. Overall, for HTIR data, the prior-surface-knowledge-optimized ANN model demonstrated superior robustness compared to the SW-ANN framework.

D. Limitations and Prospects

This article addresses HTIR remote sensing data by integrating the SW-like method with PCA transformation to construct an ANN model. This model enables direct estimation of LST, LSE, and atmospheric parameters from TOA radiance. First, the study estimated ground leaving-brightness temperature using the SW-like method. This approach enriches the model with additional land surface information beyond satellite-level data. Subsequently, PCA transformation was applied to reduce the dimensionality of spectra, simplifying the model structure while preserving spectral shape recovery. Results demonstrated that the proposed method can simultaneously retrieve the surface and atmospheric parameters with high accuracy.

However, the proposed method still has several limitations. First, the construction of the ANN model relies on simulated data, in which atmospheric profiles and emissivity spectral libraries are essential inputs. Moreover, during model development, PCA transformation was employed to reduce the dimensionality of spectral data for emissivity and atmospheric parameters. In this process, feature vectors and principal components are typically derived from fitting the sample library. For these processes, discrepancies between the library and actual conditions, especially due to factors such as mixed pixels, may introduce errors when building the retrieval model and reconstructing spectral features. Therefore, enriching the sample library to better represent real scenarios, and thereby improving the accuracy of model estimation and data dimensionality reduction, remains one of the key issues toward enhancing retrieval accuracy for HTIR data.

Additionally, the SW-like method was employed to provide prior surface information for the ANN model. This approach enables the extraction of surface information directly from the observed data without relying on external datasets, thereby offering high practicality. Our analysis indicates that incorporating surface information is beneficial for the retrieval of LST and LSE. However, the provided prior information contains uncertainties and exhibits varying error levels across different channels, which will influence the final retrieval performance. Therefore, improving the accuracy of surface prior knowledge, or introducing more robust constraints or regularization strategies within the model, is expected to further enhance the accuracy and stability of the retrieval results.

VI. Conclusion

This study proposes an ANN method with prior surface knowledge for retrieving LST, LSE, and atmospheric parameters simultaneously from HTIR remote sensing data. The findings showed that the retrieved LST achieved an accuracy of 1.49 K, LSE reached 0.011, and the model effectively preserves the characteristics of the emissivity spectra. The accuracy of atmospheric transmittivity, atmospheric upwelling radiance, and atmospheric downwelling radiance reached 0.018, 0.152, and 0.221 W cm^{-2} μm^{-1} sr^{-1}, respectively. This indicated satisfactory accuracy and confirmed the robustness of the proposed framework. Additionally, the model performed satisfactorily on real CrIS satellite data. Cross-validation results combining VIIRS LST products and AIRS LSE products indicated high consistency between the estimated parameters and the products. This confirmed the retrieval model is feasible for real satellite HTIR imagery. Furthermore, analysis indicated that incorporating prior surface information effectively enhanced the retrieval accuracy in LST, while PCA transformation improved the accuracy and spectral shape of LSE and atmospheric parameters. Therefore, integrating existing mathematical methods can provide effective knowledge guidance for ML approaches, thereby further enhancing the robustness of ML models.

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